

TERESA: Whole-Body Active Perception for Legged Robot-Assisted Emergency Assessment

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Abstract—Deploying legged mobile manipulators for autonomous emergency assessment requires the robot to actively adapt its body posture to the perceptual demands of each mission phase—locating, approaching, scanning, and performing ultrasound on an unknown patient.

This paper presents Trustworthy Emergency Robot for Efficient Scanning and Assistance (TERESA), an autonomous emergency assessment system on the Boston Dynamics Spot quadruped with a Unitree Z1 arm (hereafter Spotzi). TERESA treats body posture as an active perceptual resource, adapting it to the demands of four sequential phases: (1) the robot searches for the patient by walking forward while sweeping body posture; (2) it approaches while concurrently scanning the torso, eliminating idle time; (3) it executes an exposure scan covering the body surface and detecting injuries; and (4) it performs Focused Assessment with Sonography in Trauma (FAST) ultrasound with pre-planned posture optimisation for each anatomical target. A web-based interface allows the remote operator to click any scanned body point and trigger a full-body reconfiguration to revisit it.

Experiments on the Spotzi platform in a laboratory environment, with a medical training manikin, demonstrate fully autonomous execution of all four phases.

Index Terms—Whole-body active perception, legged manipulation, medical robotics, emergency assessment, mobile manipulation, body reconfiguration.

I. INTRODUCTION

EMERGENCY assessment protocols—including the Airway, Breathing, Circulation, Disability, Exposure (ABCDE) survey [1] and Focused Assessment with Sonography in Trauma (FAST) ultrasound [2]—enable rapid triage of life-threatening conditions at the point of care. In mass-casualty incidents and disaster sites, however, trained personnel are often unavailable, creating a critical bottleneck.

Legged platforms such as the Boston Dynamics Spot can traverse stairs, rubble, and confined spaces inaccessible to wheeled robots, and can therefore reach patients wherever they lie. Navigation alone, however, is insufficient: emergency assessment requires physical interaction with the patient—checking airway patency and respiratory effort, inspecting the skin for signs of poor circulation, scanning the body surface for wounds, burns, and haemorrhages, and performing ultrasound at specific anatomical targets. A manipulator arm extends the robot’s reachable workspace, enables multiple viewpoints of the same body region, and provides the dexterity needed for compliant probe contact.

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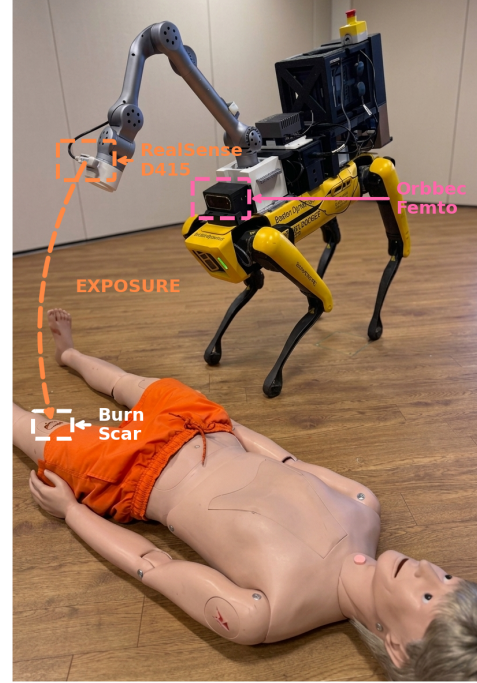


Fig. 1. The TERESA system during the exposure scanning phase: Spot quadruped (left) with Z1 arm carrying a wrist-mounted RealSense D415 for close-range torso tracking and a body-mounted Orbbec Femto Bolt for wide-field patient detection.

Robot-assisted ultrasound has advanced over two decades on fixed-base platforms. Comprehensive surveys [3], [4] document the evolution from teleoperated systems to autonomous scanning, and Hidalgo et al. [5] review current clinical applications. Key technical advances include active-sensing end-effectors [6] and autonomous protocol execution [7], [8].

The Exposure phase of ABCDE requires systematic visual inspection of the body surface for wounds, burns, and haemorrhages. Open-vocabulary detectors such as Grounding DINO [9] enable zero-shot injury detection without task-specific training data. Combined with depth back-projection from the RGB-D stream, these models produce metric 3D injury locations. To plan the exposure scan grid and the FAST the robot must first estimate the patient’s 3D body pose. Neural Localizer Fields (NLF) [10] regress 24-joint Skinned Multi-Person Linear (SMPL) skeletons from monocular RGB, providing full anatomical coverage beyond the standard 17-keypoint Common Objects in Context (COCO) format. Despite this progress, all deployed systems share a fundamental limitation: the patient must be pre-positioned within the arm’s reachable workspace, confining autonomous ultrasound and

scanning to controlled clinical settings. A possible solution is achieved by the modern Whole-Body Control (WBC). This builds on operational space formulation [11], manipulability analysis [12], and optimisation-based control [13]. These principles have been extended to legged manipulators through hierarchical Model Predictive Control (MPC) [14], Quadratic Program (QP)-based formulations [15], holistic [16] and Spot-specific grasping and manipulability enhancement [17], [18]. Across this body of work, however, the manipulation target is assumed known or static; no formulation accounts for the coupling between body posture and the quality of perceptual data used to localise that target in the first place.

Active perception research has established that deliberate sensor motion systematically outperforms passive observation [19]–[21]. This insight has been operationalised through perception-aware planning [22], [23], next-best-view grasping [24], view planning for pose estimation [25], and active neural sensing [26]. When applied to mobile manipulation, whole-body interaction [27], coupled perception-manipulation [28] and one-shot multimodal frameworks [29] have demonstrated trajectory optimisation for information gain. Critically, all of these systems operate on wheeled bases whose body posture is fixed: the base provides mobility but does not participate in shaping the perceptual field. Conversely, work on posture optimisation for legged systems—through reachability analysis [30], [31], base placement [32], body-ground contact [33], and the broader legged manipulation literature [34]—treats perception as a static input to the motion planner rather than as a quantity to be optimised. No existing system closes the perception-action loop at the whole-body level on a legged platform for medical tasks.

We advocate a paradigm in which body posture on a legged platform is treated as a first-class perceptual resource: the robot actively adapts its height, pitch, and yaw to maximise the quality of its own perceptual input, rather than solely to minimise kinematic error. The key idea is to close the perception-action loop at the whole-body level through confidence-gated search and anticipatory scanning.

This paper presents Trustworthy Emergency Robot for Efficient Scanning and Assistance (TERESA) (Fig. 1), an integrated system for autonomous emergency assessment on the Boston Dynamics Spot legged robot equipped with a Unitree Z1 manipulator arm — Spotzi. Building on the fixed-base ultrasound scanning competence established in [8], TERESA extends autonomous assessment to a fully mobile platform. We target the Exposure and FAST phases of ABCDE because they share the same hardware configuration and impose nearly identical whole-body coordination demands. TERESA addresses this through a unified whole-body active perception framework that treats body posture as an explicit control resource across all four phases.

The key contributions of this paper are as follows.

- 1) **Whole-body active perception framework** in which body posture is optimised to improve perceptual quality rather than solely to minimise kinematic error (Section III).
- 2) **Confidence-gated search with dual-sensor locking** that combines forward base search with body pitch

sweep, ensuring robust patient localisation before committing to navigation.

- 3) **Exposure body scanning pipeline** integrating an adaptive skeleton-keypoint grid with zero-shot Grounding DINO injury detection and an interactive web-based review interface (Sections III and IV).
- 4) **Pre-planned body reconfiguration** that searches over body height, pitch, and base displacement to place each anatomical target in the arm’s preferred manipulation region (Section III-C1).
- 5) **Experimental validation** on the Spotzi platform demonstrating fully autonomous execution of all four phases (Section V).

To the best of our knowledge, TERESA is the first legged robotic system to demonstrate perception-driven whole-body reconfiguration for autonomous emergency assessment.

The remainder is organised as follows: Section II formulates the problem, Section III presents the whole-body active perception framework, Section IV details the system architecture and mission Finite State Machine (FSM), Section V reports experimental results, and Section VI concludes.

II. PROBLEM STATEMENT

The Spotzi platform consists of a Boston Dynamics Spot quadruped and a Unitree Z1 six-Degree of Freedom (DOF) arm rigidly mounted on the robot’s back. Four frames are defined: the world frame $\mathcal{F}_W$ (coincident with Spot odometry), the body frame $\mathcal{F}_B$, the arm base frame $\mathcal{F}_0$ (fixed relative to $\mathcal{F}_B$), and the end-effector frame $\mathcal{F}_{EE}$ (at the Z1 flange). Two RGB-D cameras cover complementary perceptual roles: a body-mounted wide-field camera for patient detection and an End-Effector (EE)-mounted camera for high-detail torso inspection. The Spot base is modelled as a non-holonomic vehicle with configuration $\xi = [x, y, \theta]^\top \in \mathbb{R}^2 \times \mathbb{S}^1$, constrained to forward velocity v_x and yaw rate ω_z :

$$\dot{\xi} = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix} v_x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \omega_z. \quad (1)$$

The body posture is parametrised by height $h \in [h_{\min}, h_{\max}]$ and pitch $\varphi \in [0, \varphi_{\max}]$, commanded through the native posture control interface. The Z1 arm has $n = 6$ revolute joints $\mathbf{q} \in \mathbb{R}^6$ with geometric Jacobian $\mathbf{J}_{\text{arm}}(\mathbf{q}) \in \mathbb{R}^{6 \times 6}$. The full control vector is $\mathbf{u} = [\dot{\mathbf{q}}^\top, v_x, \omega_z, h, \varphi]^\top \in \mathbb{R}^{10}$, where $\dot{\mathbf{q}}$ are the arm joint velocities, v_x the forward velocity, ω_z the yaw rate, h the body height, and φ the body pitch.

The mission consists of four sequential phases, each defining a specific control subproblem.

- **Searching.** The robot must locate a stationary patient whose position is unknown a priori, using only onboard RGB-D cameras, by walking forward while adjusting body pitch, until a vision-based patient detector yields confidence above a threshold τ . The associated control problem is: find (v_x, h, φ) such that detection confidence exceeds τ , ensuring robust localisation before committing to navigation.
- **Approaching.** Once the patient is localised, the robot navigates to a pre-scanning base configuration from

which all subsequent anatomical targets are kinematically reachable. The control problem is: given the estimated patient position, find (v_x, ω_z) to reach a configuration from which every target $\mathbf{T}_i^* \in \mathcal{T}$ is reachable.

- **Exposure.** The robot performs a systematic visual inspection of the body surface, acquiring multi-view RGB-D data over an adaptive grid of anatomical keypoints to detect wounds, burns, and haemorrhages via open-vocabulary object detection. The control problem is: given the patient body model and the keypoint grid, find $(h, \varphi, \dot{\mathbf{q}})$ to scan all keypoints with sufficient viewpoint coverage and to run injury detection at each grid point.
- **FAST.** The robot acquires ultrasound images at N pre-defined anatomical target poses $\mathcal{T} = \{\mathbf{T}_i^* \in SE(3)\}_{i=1}^N$. The control problem is: given $\mathcal{T}$, find $(h, \varphi, \dot{\mathbf{q}})$ to bring the EE to each $\mathbf{T}_i^*$ with bounded error $\|\mathbf{p}_{EE} - \mathbf{p}_i^*\| \leq \varepsilon_p$, while maintaining adequate body posture for arm manipulability.

III. WHOLE-BODY ACTIVE PERCEPTION AND CONTROL

The term *whole-body* here denotes a perceptual extension of the classical WBC principle. Whereas traditional WBC coordinates all degrees of freedom to minimise kinematic error, TERESA extends this coordination to the perceptual domain: body posture and arm motion are treated as explicit control resources that together shape the robot's sensory field. By commanding posture to improve what the robot sees, from closer range and from more informative viewpoints, TERESA elevates the legged base from a passive transport platform to an active participant in the perception-action loop. This principle underpins every phase of the mission, from searching through approaching to scanning.

A. Searching

Before the robot can approach the patient, it must first localise them reliably. In an unstructured emergency scenario, the patient's position is unknown a priori. The EE-mounted camera benefits from the Z1 arm's full six-DOF mobility, which can orient the camera in any direction without moving the base. The body-mounted camera, however, has a fixed orientation relative to Spot: its downward field of view depends entirely on body pitch, making pitch the only posture parameter that directly improves its detection of a patient lying on the ground. TERESA therefore avoids time-consuming base rotations and instead searches by walking forward while sweeping body pitch.

1) *Forward search with pitch sweep:* The robot walks forward at low speed while sweeping body pitch through a sequence of angles $\varphi \in [0, \varphi_{\max}]$. The body-mounted camera runs YOLO11-pose and a posture classifier, outputting a confidence score $c \in [0, 1]$; the EE-mounted camera runs a torso tracker that scans the surroundings independently of the base heading. Algorithm 1 formalises the confidence-gated search loop. This strategy exploits the arm's native mobility for rotational search while using body pitch—the one viewing parameter the arm cannot compensate for—to optimise the body-mounted camera's fixed viewpoint. When

a lock is confirmed (Algorithm 1), the arm returns home and the coordinator collects N approach-point samples in $\mathcal{F}_W$, computing their mean as the navigation target, while the NLF pipeline accumulates skeleton estimates with exponential moving average smoothing. If the detection is temporarily lost, the system tolerates a brief absence before resuming the search.

Algorithm 1 Confidence-Gated Search with Dual-Sensor Lock

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1: Input:  $\varphi_{\max}$ , thresholds  $\tau_{\text{high}}, \tau_{\text{low}}$ ,  $N$  samples
2: Output: navigation target  $\mathbf{p}_{\text{nav}} \in \mathbb{R}^3$ , skeleton estimate
3: while not locked do
4:   for  $\varphi \in \{\varphi_1, \dots, \varphi_{\max}\}$  do
5:      $c_b \leftarrow \text{BODYDETECT}(\varphi)$ 
6:     if  $c_b > \tau_{\text{high}}$  then
7:        $\text{ARMHOME}()$ 
8:       for  $i = 1$  to  $N$  do
9:          $\mathbf{p}_i \leftarrow \text{GETAPPROACHPOINT}()$ 
10:         $\text{NLFACCUMULATE}(\mathbf{p}_i)$ 
11:      end for
12:       $\mathbf{p}_{\text{nav}} \leftarrow \text{MEAN}(\{\mathbf{p}_i\})$ 
13:    end for
14:    return lock
15:  end if
16:   $c_e \leftarrow \text{EEDTECT}()$ 
17:  if  $c_e > \tau_{\text{low}}$  then
18:     $\text{WALKTOWARD}(\text{detection})$ 
19:     $\text{PITCHREFINE}(\varphi)$ 
20:  end if
21: end for
22:  $\text{ADVANCEFORWARD}()$ 
23: end while

```

B. Approaching

In traditional mobile manipulation pipelines, the robot first navigates to a target, stops, and only then begins torso mapping—introducing significant idle time. TERESA eliminates this idle period by mapping the torso concurrently with the approach. Algorithm 2 formalises the parallel base-controller / torso-mapper execution.

The arm controller orients the EE-mounted camera toward the patient's torso centroid. Joint velocities are computed via the damped pseudo-inverse of the arm Jacobian:

$$\dot{\mathbf{q}} = \mathbf{J}_{\text{arm}}^\top (\mathbf{J}_{\text{arm}} \mathbf{J}_{\text{arm}}^\top + \mu \mathbf{I}_6)^{-1} \mathbf{v}_{\text{des}}, \quad (2)$$

where $\mu > 0$ prevents singularities near kinematic limits. Only the angular component of $\mathbf{v}_{\text{des}}$ is active: it encodes the look-at rotation $\mathbf{x}_{EE} = (\mathbf{p}^* - \mathbf{p}_{EE}) / \|\mathbf{p}^* - \mathbf{p}_{EE}\|$, keeping the torso centred in the frame; the translational component is zero, holding a nominal home position. The base approach speed v_x is modulated by the detection confidence: the robot slows when the torso estimate is uncertain and resumes full speed once confidence is restored. We refer to this confidence-gated modulation as the QualityMonitor. At each viewpoint, depth and skeleton data are fused into 3D keypoint estimates via a weighted average of detection rate and spatial stability. From the fused torso model, the five canonical FAST target poses are computed as fixed offsets from the torso centroid in $\mathcal{F}_0$.

Algorithm 2 Anticipatory Approaching with Concurrent Torso Mapping

```

1: Input:  $\mathbf{p}_{\text{nav}} \in \mathcal{F}_W$ , torso keypoint confidences
2: Output: fused torso model, FAST target poses  $\mathcal{T}$ 
3: parallel do
4:   // Base navigation
5:   while  $\|\mathbf{p}_{\text{nav}} - \boldsymbol{\xi}\| > \varepsilon$  do
6:      $v_x, \omega_z \leftarrow \text{PROPORTIONALCTRL}(\mathbf{p}_{\text{nav}}, \boldsymbol{\xi})$ 
7:      $v_x \leftarrow \text{QUALITYMONITOR}(v_x, \text{confidence})$ 
8:   end while
9:   // Torso mapping
10:  while not handoff do
11:     $\hat{\mathbf{q}} \leftarrow \text{DAMPEDPINV}(\mathbf{J}_{\text{arm}}, \mathbf{v}_{\text{des}})$  ▷ Eq. 2
12:    if all 4 keypoints confident then
13:      grid  $\leftarrow$  2-pose minimal pattern
14:    else
15:      grid  $\leftarrow$  3-pose comprehensive pattern
16:    end if
17:    for each pose  $\in$  grid do
18:       $\text{FUSEDDEPTH}(\text{pose})$ 
19:    end for
20:  end while
21: end parallel
22:  $\mathcal{T} \leftarrow \text{FASTTARGETOFFSETS}(\text{fused\_torso})$ 

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C. Exposure

After the torso mapper has localised the patient’s anatomy, Phase 3 executes a systematic visual inspection of the body surface, serving the “E” (Exposure) component of the ABCDE protocol. A key design principle is that the remote operator remains in the loop: the system automates the scanning motion but preserves the clinician’s ability to inspect, re-examine, and override.

1) *Posture optimisation for body reconfiguration:* Once the anticipatory torso mapper has localised the patient’s anatomy and computed the target set $\mathcal{T}$ (both FAST ultrasound targets and exposure grid points) in $\mathcal{F}_W$, the system pre-plans an optimal body posture for each target through a two-tier kinematically-aware stance search with Inverse Kinematics (IK)-driven retry. This optimisation is shared with Phase 4 (Section III-D).

The posture optimisation space spans three parameters: body height $h \in [h_{\min}, h_{\max}]$ and pitch $\varphi \in [0, \varphi_{\max}]$, as defined in Section II, plus a body-frame base displacement $dy \in \mathbb{R}$ along the patient’s head-to-foot axis, expressed in the patient-aligned body frame (Section IV). The discrete search grids are $\mathcal{H} = \{h_1, \dots, h_{N_h}\}$ and $\mathcal{P} = \{\varphi_1, \dots, \varphi_{N_p}\}$, with dy sampled uniformly at 0.10 m intervals. The cost function includes a regularisation term that penalises lateral displacement:

$$J(h, \varphi, dy) = \|\mathbf{T}_{\mathcal{F}_W}^{\mathcal{F}_0}(h, \varphi, dy) \mathbf{p}^* - \mathbf{p}_{\text{sweet}}\| + \lambda |dy|, \quad (3)$$

where $\mathbf{p}_{\text{sweet}} \in \mathbb{R}^3$ is the empirically determined centre of the Z1 arm’s preferred manipulation region in $\mathcal{F}_0$, and $\lambda = 0.50$ weights lateral movement against reachability. The value $\lambda = 0.50$ was chosen empirically to balance reachability against energy cost: $\lambda > 1.0$ prevented necessary base displacement, while $\lambda < 0.25$ caused excessive walking for marginal reachability gains. This penalty encourages the optimiser to prefer height and pitch adjustments over base walking—reducing

energy consumption and producing smoother, more predictable motion—while still allowing longitudinal displacement when strictly necessary. The two-tier escalation proceeds as follows:

- 1) **In-place search**— (h, φ) only: a coarse grid sweeps $\mathcal{H} \times \mathcal{P}$ ($N_h N_p$ combinations). The posture yielding an IK solution within 2 s is selected.
- 2) **Base-displacement search**— (dy, h, φ) : when the in-place search fails, the optimiser adds longitudinal base displacement dy , expanding to $N_{dy} N_h N_p$ combinations. If the IK solver times out, the point is skipped.

The optimisation is computed once per target after $\mathcal{T}$ is available, imposing no real-time computational burden. Body reconfiguration is executed between targets: Spot assumes the pre-computed posture (h_i^*, φ_i^*) , settles, and the base navigator drives the dy displacement before the Z1 arm executes the IK trajectory toward the transformed target.

2) *Posture-adaptive scan grid:* The scan grid adapts to the patient’s estimated body pose. Four torso keypoints—left and right shoulders, left and right hips—define a bilinear patch over the body surface. Let $\mathbf{p}_{SL}, \mathbf{p}_{SR}, \mathbf{p}_{HL}, \mathbf{p}_{HR} \in \mathbb{R}^3$ denote these keypoints in $\mathcal{F}_W$. The grid is generated by interpolating uniformly along the two parametric directions: head-to-foot (u -axis, from shoulders to hips) and lateral (v -axis, from left to right). A grid point at parameters $(u, v) \in [0, 1] \times [0, 1]$ is computed as $\mathbf{p}(u, v) = (1-u)(1-v) \mathbf{p}_{SL} + u(1-v) \mathbf{p}_{HL} + (1-u)v \mathbf{p}_{SR} + uv \mathbf{p}_{HR}$, with a fixed height offset to maintain a safe camera-to-skin distance. Typical grid resolution is 0.15 m in both directions, producing 15–25 points depending on torso dimensions. When the NLF prior is available, all 24 SMPL joints are populated, extending the grid to the head, hands, and feet for full-body coverage. Grid points are visited in a serpentine path to minimise arm travel.

3) *Injury detection:* At each grid point, Grounding DINO [9]—a zero-shot open-vocabulary object detector—runs on the EE-mounted camera RGB stream at approximately 1 fps on the Jetson Orin GPU, prompted with a vocabulary of wound-related terms (wound, laceration, bleeding, burn, etc.). We select Grounding DINO over alternatives such as OWL-ViT [35] and GLIP [36] because its joint visual-language grounding produces tighter bounding boxes for fine-grained injury descriptions, which is critical for 3D back-projection accuracy. Detections are projected to 3D world coordinates via the registered depth channel and associated with their grid point index. This 3D projection is required because the operator’s clicks during interactive review trigger a physical robot repositioning to the corresponding body surface point, which demands metric 3D coordinates.

4) *Web-based interactive review:* Following the sweep, the web interface (Section IV) overlays the scan grid on the EE-mounted camera feed with detected injuries highlighted as colour-coded bounding boxes. The remote operator can click any grid point to command a full re-inspection of that body region.

D. FAST

Phase 4 performs autonomous ultrasound acquisition at the N FAST targets estimated during Phase 2. Body posture

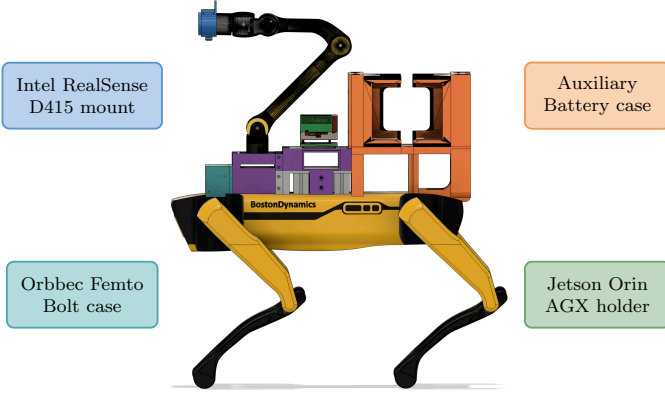


Fig. 2. Mechanical integration of the Z1 arm, Orbbec Femto Bolt, RealSense D415, and auxiliary battery onto Spot's payload frame.

optimisation for each $\mathbf{T}_i^* \in \mathcal{T}$ follows the same two-tier procedure described in Section III-C1. The critical challenge of this phase is compliant probe contact: the ultrasound probe must maintain controlled pressure against the patient's skin—excessive force risks injury, while insufficient contact loses the acoustic signal. On a legged platform, this is compounded by the fact that the Z1 arm provides no native impedance control, requiring us to implement a custom torque-level controller for safe physical interaction.

1) *Contact control*: Ultrasound scanning requires compliant probe contact. To achieve this on our platform, TERESA implements a custom Cartesian impedance controller on the Z1 torque interface at 500 Hz. The desired EE pose is computed relative to the torso surface centroid $\mathbf{p}_s$ and outward normal $\mathbf{n}_s$, placing the probe at $\mathbf{p}_{des} = \mathbf{p}_s + d\mathbf{n}_s$, where d is the desired contact offset. Joint torques are computed as

$$\boldsymbol{\tau} = \mathbf{J}_{arm}(\mathbf{q})^\top [\mathbf{K}_p(\mathbf{p}_{des} - \mathbf{p}_{EE}) - \mathbf{K}_d \dot{\mathbf{p}}_{EE}] + \mathbf{g}(\mathbf{q}) + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}), \quad (4)$$

where $\mathbf{K}_p, \mathbf{K}_d \in \mathbb{R}^{3 \times 3}$ are adaptive stiffness and damping gains that scale linearly with arm extension to prevent overloading at the workspace boundary, $\mathbf{g}(\mathbf{q})$ and $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ are the gravity and Coriolis terms computed via Pinocchio [37], and $\dot{\mathbf{p}}_{EE} = \mathbf{J}_{arm}(\mathbf{q})\dot{\mathbf{q}}$ is the end-effector velocity. A safe switching service alternates between the impedance controller and a position-controlled joint trajectory controller, defaulting to position control on error.

IV. SYSTEM ARCHITECTURE

The TERESA system integrates heterogeneous hardware and software components over Robot Operating System (ROS) 2 with Data Distribution Service (DDS). This section describes how these components are connected into a deployable field system, from the physical platform through the coordination layer to the remote monitoring interface.

A. Hardware Platform

The TERESA hardware platform builds on *Spotzi*. To this base configuration we added an Orbbec Femto Bolt body-mounted RGB-D camera for wide-field patient detection, an

Intel RealSense D415 wrist-mounted at the Z1 end-effector for close-range torso inspection, an auxiliary battery pack that powers the Z1 and the Orbbec independently of Spot's internal supply, and an NVIDIA Jetson Orin AGX for onboard perception and control. The entire assembly was designed in Fusion 360 (Fig. 2) with a modular bracket system that allows components to be easily swapped, removed, or extended without modifying the underlying structure. The centre of mass is placed slightly rearward of Spot's geometric centre to counterbalance the arm's forward displacement when extended toward the patient. Several candidate arms were evaluated within the Fusion 360 assembly; the Z1 was selected for its light weight and compact form factor, which preserve Spot's payload margin for the onboard computer and sensors while avoiding interference with the base during locomotion. Spot's five built-in stereo cameras are positioned low on the body for ground-plane obstacle avoidance and are not suitable for skeleton detection. Computation is distributed across two units: Spot's onboard computer handles base control and odometry, while an NVIDIA Jetson Orin AGX runs perception, arm control, task coordination, and WBC via Docker containers with CUDA 11.4.

B. Software Architecture

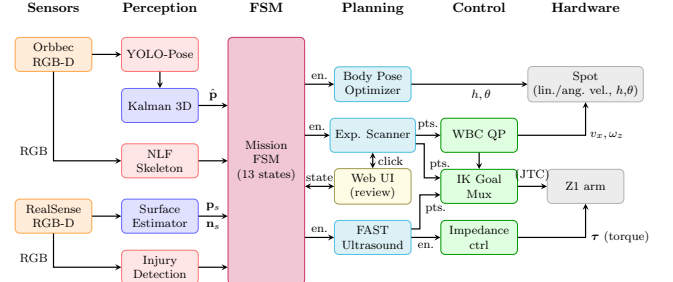


Fig. 3. System block diagram. The architecture spans sensors, perception, FSM, planning, control, and hardware. Dashed arrows denote operator-triggered or posture-optimisation paths.

Fig. 3 shows the complete system architecture. The Orbbec and RealSense RGB-D streams feed the Perception layer, which runs YOLO-pose skeleton detection, Kalman-filtered 3D back-projection, NLF 3D body pose estimation, surface estimation, and Grounding DINO injury detection. Perception outputs route to the 13-state FSM, which enables four Planning modules: the Exposure Scanner, the FAST Ultrasound planner, the Body Pose Optimizer, and the Web UI for interactive review. The Control layer translates plans into actuation: the WBC QP publishes base velocity $[v_x, \omega_z]$ to Spot and arm goals to the IK Goal Multiplexer; the Impedance Controller manages compliant contact during FAST scanning, publishing joint torques $\boldsymbol{\tau}$ directly to the Z1. The Z1 FSM manages the arm's internal state transitions—homing, look-at tracking, scanning trajectory execution, and impedance-controlled contact—and synchronises with the WBC coordinator. Dashed arrows in the diagram denote operator-triggered or posture-

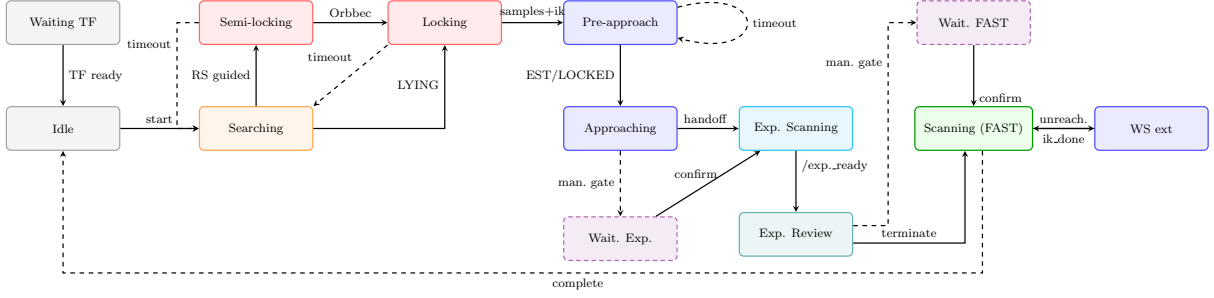


Fig. 4. FSM with 13 states. Dashed transitions are perception-driven; solid transitions are geometry-driven.

optimisation paths. An emergency stop immediately halts the base and returns the arm to a safe configuration.

C. Mission Management

The four assessment phases of Section II are orchestrated by a 13-state FSM (Fig. 4). After Waiting TF and Idle ensure system readiness, the FSM advances through Searching (Phase 1), Approaching (Phase 2), Exposure (Phase 3), and FAST (Phase 4). Perception confidence triggers dashed transitions; geometric conditions drive solid edges. Timeouts at any point revert to the search, preventing commitment to false detections. The FSM serves as a pragmatic integration layer that sequences the methods of Section III into a deployable field system; alternative orchestration paradigms could extend this approach in future work.

D. Remote Monitoring

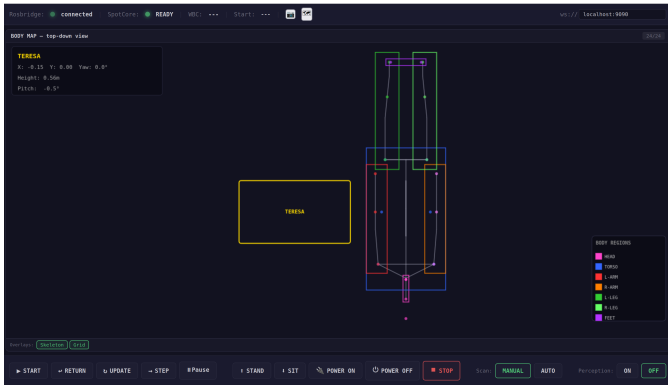


Fig. 5. Web-based operator interface during the exposure scanning phase. The Body Map (top-down view) shows the patient mannequin skeleton with colour-coded body regions, the exposure scan grid points, and the Spot robot position (yellow). The info panel (top-left) displays Spot's current posture parameters. During exposure review, the operator can click any grid marker to command the robot to re-inspect that body region.

Full autonomy in emergency assessment is neither feasible nor desirable: the remote operator must retain the ability to override or re-inspect regions of clinical interest. TERESA bridges this gap with a web interface that displays a top-down view of the patient mannequin and scan grid—the Body Map.

A single click on any grid marker in this map translates into a 3D whole-body repositioning command (posture optimisation, base navigation, and arm trajectory replay) without manual teleoperation.

The TERESA web interface provides dual-camera monitoring and robot control via a browser-based dashboard. During exposure review, the RealSense overlay renders the scan grid as clickable markers; each click triggers a full re-positioning sequence—posture optimisation, settle, and IK trajectory replay—within the same ROS 2 topic protocol used for autonomous operation. A manual/automatic scan gate toggle allows the operator to require explicit confirmation before advancing between phases. All web commands are mirrored on the keyboard controller for redundancy. Fig. 5 shows the interface during review.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

Experiments ran on the Spotzi platform with an Orbbec Femto Bolt on Spot's body and a RealSense D415 at the EE. All neural network inference runs on an NVIDIA Jetson Orin AGX (ROS 2 Humble, CUDA 11.4, PyTorch 2.0): YOLO11n-pose at 23–30 fps, NLF at 13–17 fps, Grounding DINO at 1–2 fps. Spot's integrated computer handles base control and odometry.

Ten supine mannequin positions were defined (2 m–4 m, $\pm 45^\circ$); each was tested in two configurations, yielding twenty paired end-to-end trials. We also conducted a separate search ablation with twenty additional trials at five orientations (0° – 180°), comparing the full TERESA system against a Spot-only baseline with the arm disabled.

The two end-to-end configurations differ in three aspects. **Baseline** uses a conventional sequential pipeline: navigation, torso mapping, and scanning execute in strict order with no concurrency; body posture is fixed at Spot's nominal stance; the exposure scan uses a fixed three-pose pattern; body reconfiguration is disabled. **TERESA** adds anticipatory torso mapping concurrent with approach (QualityMonitor, Section III-B), an adaptive exposure grid (two or three poses depending on keypoint confidence), and pre-planned two-tier body reconfiguration (Section III-C1) for every FAST target.

We evaluate five metrics: *total completion time* T_{total} (wall-clock from start to Idle); *lock time* T_{lock} (time to confirmed

patient lock); *scan overlap* $\eta = N_{\text{pre}}/N_{\text{total}}$ (fraction of torso model completed during approach; a value near 1 means mapping imposed no extra time); *idle time* T_{idle} (cumulative stationary time); and *EE positioning error* $e_i = \|\mathbf{p}_{EE,i} - \mathbf{p}_i^*\|$ (Euclidean distance to each FAST target, reported as mean and standard deviation across all kinematically reachable targets).

B. End-to-End System Performance

Across all twenty end-to-end trials, TERESA reduced total completion time by 10.5% (Table I). The gain came from eliminating idle time—the anticipatory torso mapper completed $\sim 80\%$ of the torso mapping concurrently with approach—and from the adaptive grid, which selected the minimal two-pose pattern in the majority of trials. Note that the baseline benefits from an optimistic assumption: without body reconfiguration, many FAST targets are unreachable from the default posture (Section V-D). Manual operation, where operators skipped unreachable grid points, reduced total time to ~ 70 s at the cost of incomplete coverage. The principal contribution of TERESA is therefore not speed but capability: without whole-body reconfiguration, the baseline simply cannot reach many anatomical targets and returns incomplete scans with higher positioning error. TERESA achieves systematic, precise, and complete assessment across all targets, which the baseline structurally cannot deliver.

TABLE I
END-TO-END RESULTS ACROSS TWENTY PAIRED TRIALS (MEAN $\pm$ STD).

Metric	Baseline	TERESA
Total completion time (s)	100.2 $\pm$ 8.1	89.7 $\pm$ 7.3
Scan overlap η	0.00	0.80 $\pm$ 0.12
EE positioning error (mm)	44.7 $\pm$ 5.2	29.8 $\pm$ 4.1

C. Search Phase: Whole-Body Coverage and Dual-Sensor Lock

TERESA required no base rotation: the arm’s seven scanning poses provided full 360° coverage as the base walked forward, yielding a mean lock time of 3.2(11) s. The Spot-only baseline, limited to the Orbbec’s 77° field of view, needed five discrete base rotations (~ 7 s each at 0.2 rad s^{-1} , plus 2.0 s dwell), taking 14.8(23) s—a $5.0\times$ slowdown confirmed by a 10 000-trial Monte Carlo simulation.

The dual-sensor architecture added robustness independently of speed: in five TERESA trials the Orbbec alone failed to lock because its fixed horizontal viewpoint (0.5 m above ground) is suboptimal for detecting a supine body. The arm-mounted RealSense, elevated by the WBC look-at controller, detected partial torso keypoints and guided the base to a more favorable orientation for the Orbbec. Without this redundancy, these trials would have resumed the search cycle, adding ~ 14 s. The arm thus serves two complementary roles: 360° coverage eliminates the base-rotation bottleneck, while its elevated viewpoint provides sensor redundancy against single-camera detection failures.

TABLE II
SEARCH ABLATION: TERESA VS. SPOT-ONLY BASELINE (20 TRIALS, 5 ORIENTATIONS).

Metric	Spot-only	TERESA
Lock time (s)	14.8 $\pm$ 2.3	3.2 $\pm$ 1.1
Speedup	—	$5.0\times$
Base rotations required	5	0
Detection failures	5/10	0/10*

*Orbbec alone failed in 5/10 trials; the arm-mounted RealSense rescued all five by guiding the base to a better viewpoint.

D. Whole-Body Reconfiguration

Fig. 6 compares four posture configurations of increasing capability. The fixed nominal posture (Spot’s default stance) reaches only a fraction of targets. A single heuristic posture (-0.15 m height, 5° pitch) improves coverage but cannot adapt to varying target positions. The in-place optimiser selects the best (h, φ) per target via kinematically-aware stance search with IK-driven retry. The base-displacement optimiser adds longitudinal base walking dy with the regularised cost $J = \|\mathbf{p} - \mathbf{p}_{\text{sweet}}\| + \lambda |dy|$ ($\lambda = 0.50$): the penalty encourages height and pitch adjustments over walking, while still allowing displacement when necessary. The residual 15% of exposure points at the furthest extremities (head, feet) were beyond the arm’s maximum reach even with full base repositioning and were skipped gracefully.

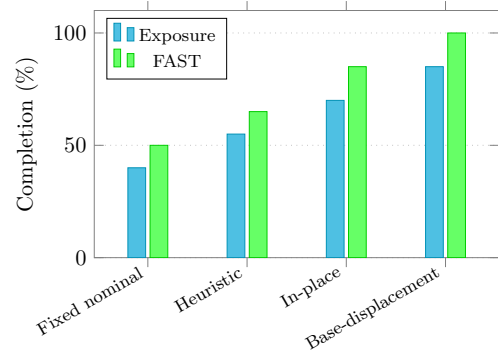


Fig. 6. Target completion rate for four body posture configurations. The FAST targets benefit more from each tier because their 5 cm accuracy requirement permits smaller posture adjustments than the exposure scan’s continuous coverage.

E. Exposure Body Scanning and Injury Detection

The exposure pipeline was evaluated with XX trials on a supine mannequin at a per-point dwell of 2 s. Table III summarises the detection performance across all trials. Grounding DINO detected wounds, burns, and hemorrhages across the adaptive scan grid with a recall of XX% and XX false positives per scan on average. Injuries were projected to 3D via the registered depth channel with a localisation error of $XX \pm XX$ mm. Fig. 7 shows example detections.

TABLE III
INJURY DETECTION PERFORMANCE ACROSS XX EXPOSURE TRIALS.

Metric	Value
Scan points per trial	XX $\pm$ XX
Scan duration (s)	XX $\pm$ XX
Wound recall (%)	XX
False positives per scan	XX
3D localisation error (mm)	XX $\pm$ XX



Fig. 7. Example injury detections from the RealSense RGB stream with Grounding DINO bounding boxes, predicted class, and confidence.

VI. CONCLUSION

This paper presented TERESA, a legged mobile manipulation system that treats body posture as an active perceptual resource, enabling a single platform to search, approach, scan, and perform ultrasound on an unknown patient.

This work shows that the manipulator arm simultaneously serves as a tool for interaction and perception—a duality amplified by legged posture. TERESA exploits this coupling through posture-perception co-design rather than explicit modelling, a practical strategy for legged robots operating beyond structured settings.

In future work, we would like to extend TERESA to multi-patient scenarios and the full ABCDE protocol. The existing platform can support the remaining components without architectural changes: airway inspection and chin-lift via the impedance controller, a digital stethoscope for breathing, a pulse oximeter for circulation, and pupillary response analysis for disability. These steps would bring autonomous emergency triage closer to field deployment.

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